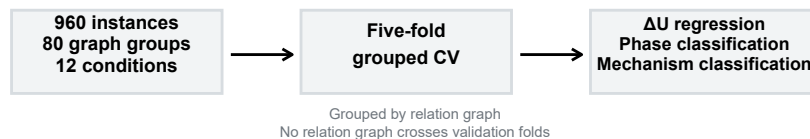


Hidden-state proxies outrank vocabulary readbacks in the controlled proxy audit

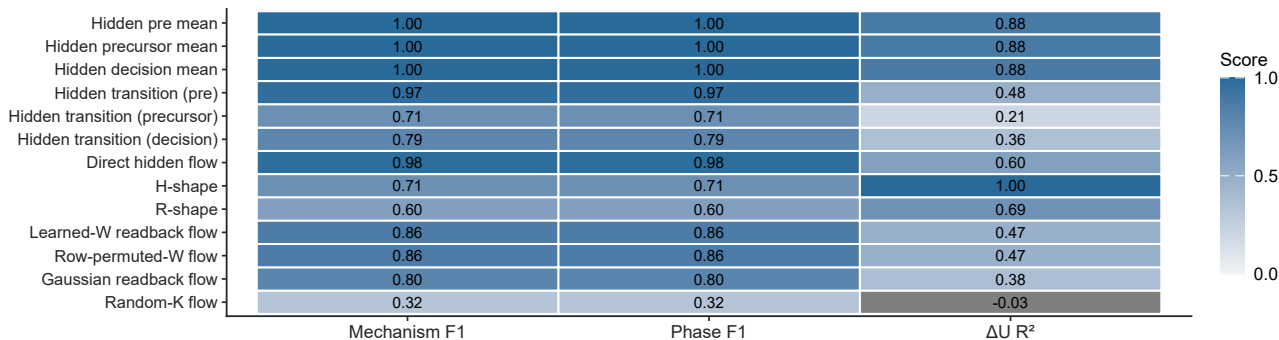
a



b

Complete proxy ranking across checkpoints

Cell values are checkpoint means; metrics remain separate

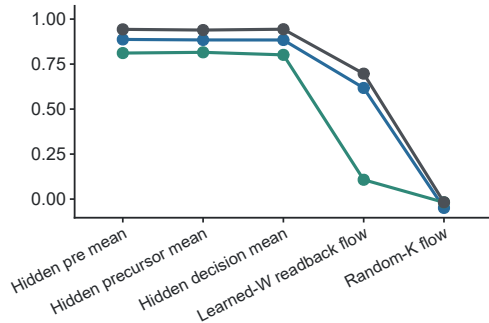


c

Initial proxy audit before capacity matching

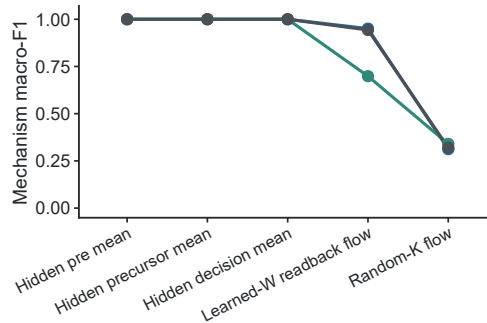
Qwen Llama Gemma

$\Delta U R^2$



Mechanism-label separation

Qwen Llama Gemma



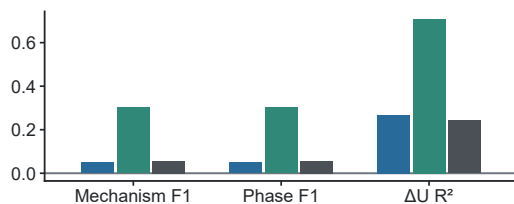
e

Hidden precursor minus learned-W readback

Positive values favour the hidden-state proxy

Qwen Llama Gemma

Metric difference



f

Object comparison updated

Ordered layer-state trajectory

Strict predecision + matched capacity
reported in Supplementary Fig. S13